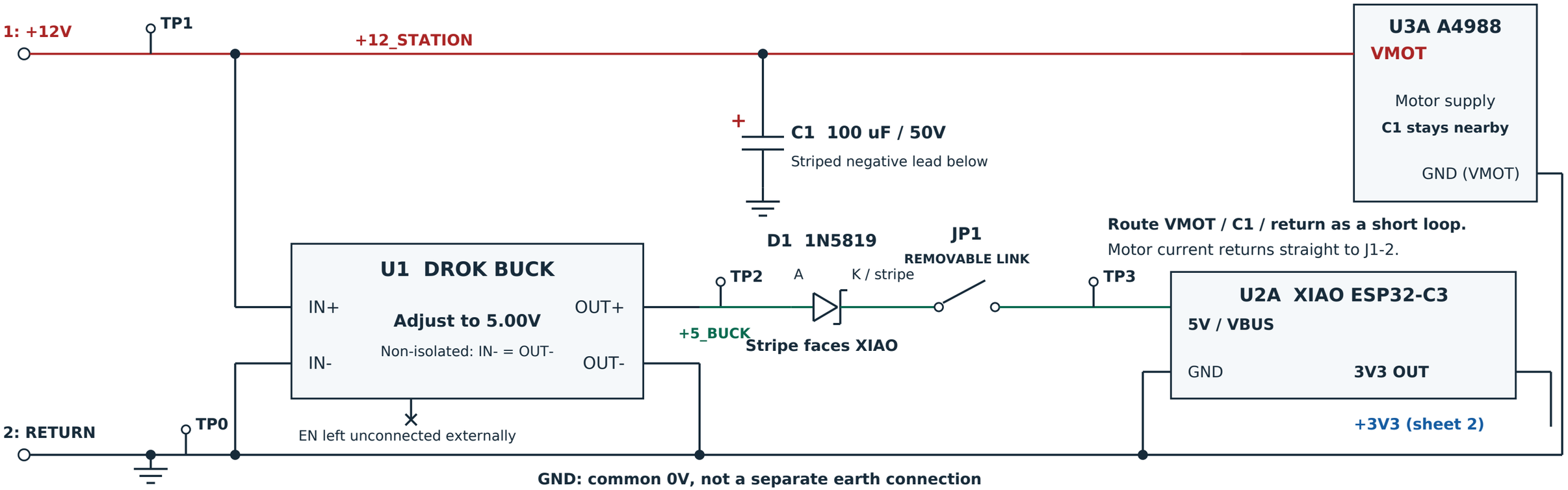
# Power circuit - one station

Real wire connections, component symbols, and named nets. Filled dot = connected junction. Same net label = same electrical node.

## J1 TWO-WIRE INPUT

Branch fuse is at supply end (see below).



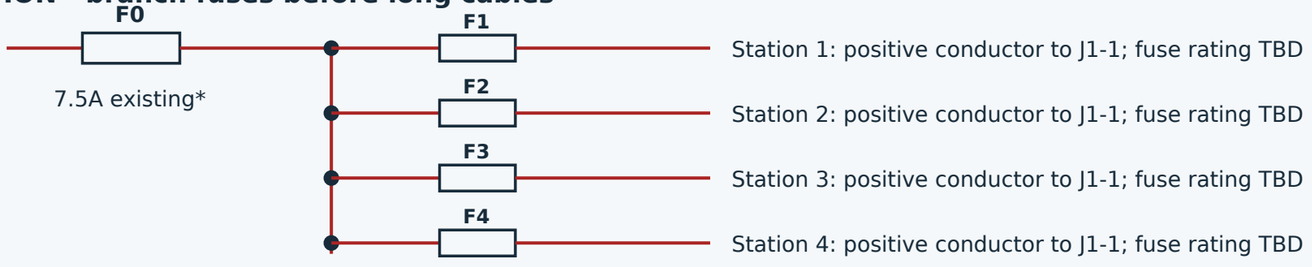

## USB / EXTERNAL POWER RULE

**USB plugged in: JP1 OPEN. Standalone from 12V: USB unplugged, JP1 CLOSED. Change JP1 only with all power off.**

The diode blocks backfeed into the buck; it is not a complete two-source power selector or 12V overvoltage protector.

### SUPPLY DISTRIBUTION - branch fuses before long cables

**ALITOVE 12V / 6A**




**Supply negative splits to** each station J1-2.

*** F0 is recorded, not approved** for unknown wire sizes.

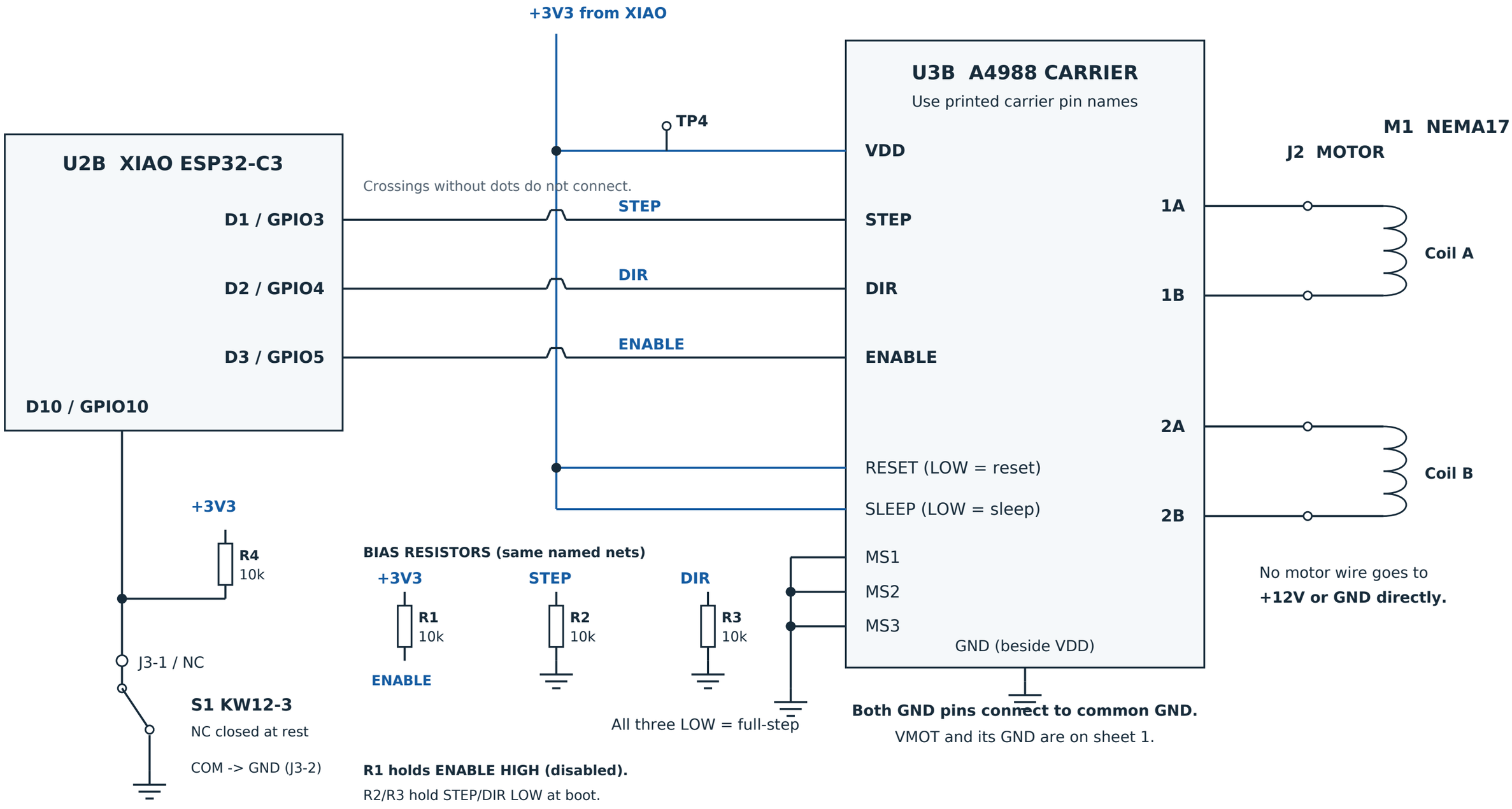
6A total capacity still needs testing.

**DESIGN REVIEW - NOT AS-BUILT CERTIFICATION. Do not reuse the suspect XIAO until isolated and assessed.**

Symbols are functional; use module pin labels, not drawing position.

# Control circuit, switch, and motor

U2B and U3B are the SAME physical boards as U2A/U3A on sheet 1. No extra ESP32 or driver is implied.




NO unused. INPUT_PULLUP in firmware.

R4 is an added external pull-up; the earlier test used only the ESP32 internal pull-up. Short/open switch detection is not equivalent:

**open wire reads STOP; a short to GND can falsely read READY. Test both switch states and unplug detection before motion.**

# Connection ledger and meter checks

This sheet is the exact connectivity reference. Net names repeat across sheets. Measure at labeled test points rather than crowded pins.

| NET / FUNCTION | ALL CONNECTED POINTS |
|---|---|
| +12_STATION | J1-1; buck IN+; A4988 VMOT; C1 positive; TP1 |
| GND / 0V | J1-2; buck IN-/OUT-; XIAO GND; both A4988 GND pins; C1 negative; TP0 |
| GND (continued) | A4988 MS1/MS2/MS3; R2/R3 lower ends; switch COM (J3-2) |
| +5_BUCK | Buck OUT+; D1 anode (unstriped); TP2 |
| DIODE OUTPUT | D1 cathode (striped); JP1 buck-side contact |
| XIAO_5V | JP1 XIAO-side contact; XIAO 5V/VBUS; TP3 (USB VBUS is on this board net) |
| +3V3 | XIAO 3V3 OUT; A4988 VDD/RESET/SLEEP; R1/R4 upper ends; TP4 |
| STEP | XIAO D1 / GPIO3; A4988 STEP; R2 upper end |
| DIR | XIAO D2 / GPIO4; A4988 DIR; R3 upper end |
| ENABLE | XIAO D3 / GPIO5; A4988 ENABLE; R1 lower end |
| HOME_NC | XIAO D10 / GPIO10; switch NC (J3-1); R4 lower end |
| MOTOR OUTPUTS | 1A/1B to one measured coil pair; 2A/2B to the other. No common rail connection. |

## POWER MODES - JP1 disconnects only the external positive feed; ground remains common.

| MODE | USB | 12V | JP1 | RESULT |
|---|---|---|---|---|
| USB logic / upload | Connected | Off | OPEN | XIAO + driver logic powered; VMOT off |
| USB motor bench test | Connected | On | OPEN | USB powers logic; 12V powers motor and unloaded buck |
| Standalone station | Unplugged | On | CLOSED | Buck powers logic; 12V powers motor |
| Not allowed in this revision | Connected | On | CLOSED | Two sources meet on VBUS; do not use this configuration |

## DC VOLTAGE CHECKS: black probe at TP0 (J1 negative), red at the selected test point.

TP1: ~12V with supply on. TP2: 5.00V before diode. TP3: ~USB voltage in USB mode, lower than TP2 through diode standalone.

TP4: ~3.3V. RESET/SLEEP: ~3.3V. ENABLE: ~3.3V idle, near 0V during jog. HOME: near 0V released, ~3.3V pressed/open.

A normal meter cannot reliably resolve the short STEP pulse or the 0.4-second ENABLE window. Use scope/logic analyzer if needed.

# Audit findings and controlled restart

Schematic reviewed against manufacturer references. Actual perfboard, motor rating, fuse sizing, and suspect components remain unverified.

## WHAT IS VERIFIED / WHAT IS NOT

| FINDING | DETAIL |
|---|---|
| Correct topology | 12V to VMOT and buck input in parallel; 3.3V to driver VDD/RESET/SLEEP; one common ground. |
| Reported measurements | VDD, RESET and SLEEP each 3.3V; VMOT-to-adjacent-GND read 0V; later XIAO heated and lost USB recognition. |
| Diagnosis not established | No proof of short location or failed part. VMOT=0 can be missing feed, missing return, fuse, supply or probe issue. |
| USB correction | JP1 OPEN for USB. External diode alone does not isolate computer VBUS from all external-power conditions. |
| Protection limitation | DROK family lists no output overvoltage or input reverse-polarity protection. D1 is not an overvoltage clamp. |
| New items shown | JP1 disconnect and R2/R3/R4 = 10k resistors are proposed additions; presence in your build is not confirmed. |
| No fixed Vref yet | I_limit = Vref / (8 x Rsense). Need actual motor phase-current rating and both driver sense-resistor markings. |
| No approved fuse value | F0=7.5A is existing stock. F1-F4 must protect branch wire/connectors; select with current and fuse-curve data. |
| Later upgrade only | Reserve XIAO D4/GPIO6 and D5/GPIO7 for encoder I2C. Exact encoder board power/pull-ups are not approved here. |

## RESTART SEQUENCE - leave the suspect XIAO disconnected throughout passive checks.

1. Remove USB and 12V. Confirm capacitor voltage has fallen near 0V before resistance tests; do not short it with a tool.
2. Isolate/remove suspect XIAO and driver. Check bare wiring against sheet 3; inspect both sides for bridges and shared copper strips.
3. Check intended continuity, including both driver ground socket contacts to J1-2. Check separate power rails for unintended shorts.
4. With electronics disconnected, test supply before/after each fuse against supply negative. Test buck alone and verify 5.00V at TP2.
5. Use a current-limited lab supply for fault finding if available. A 7.5A fuse is not fast semiconductor protection or overvoltage protection.
6. Verify a known-good XIAO separately. JP1 open: add checked logic wiring/driver incrementally, powering off before every change.
7. With EN held HIGH, verify TP1/TP4 and set Vref from confirmed motor/Rsense data. Never guess based on screw direction.
8. Power off; connect checked coil pairs. Run only the short, unloaded jog and switch test. No suspended load or room operation.

## SOURCE REFERENCES (manufacturer documentation; accessed 2026-10-16 America/Chicago)

**Seeed C3 pin map and board schematic** https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/

**A4988 carrier guidance / Allegro IC datasheet** https://www.pololu.com/product/1182 | https://www.pololu.com/file/0J450/A4988.pdf

**DROK matching module family (exact revision unverified)** https://www.droking.com/10pcs-dc-buck-converter-5v-voltage-regulator